

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2023

Ch 1213

(Chemistry II)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|---|----|
| 1(a). Deduce the rate expression for second order reaction, where both the concentration terms are same. | 09 |
| 1(b). What is energy of activation? How can energy of activation be determined with the help of Arrhenius equation? | 09 |
| 1(c). Distinguish between order and molecularity of a reaction. Illustrate the following terms:
(i) Steady state approximation, (ii) Consecutive reactions | 11 |
| 1(d). 50% of a first order reaction is completed in 50 minutes. Calculate the time required to complete 99% of the reaction. | 06 |
| 2(a). Define quantum efficiency. Why the quantum efficiency for the hydrogen and chlorine reaction is very high? | 10 |
| 2(b). What are photochemical reactions? Explain primary and secondary photochemical reaction with suitable examples. | 12 |
| 2(c). Define the following terms giving examples.
(i) Photosensitized reactions, (ii) Chemiluminescence | 06 |
| 2(d). Write down the differences between fluorescence and phosphorescence. | 07 |
| 3(a). What is geometrical isomerism. Discuss the geometrical isomerism of maleic and fumaric acid. | 10 |
| 3(b). Define optical isomerism. Briefly discuss the term “specific rotation” with example. | 10 |
| 3(c). Describe the biological importance of chirality. | 06 |
| 3(d). Discuss the reaction mechanism of cationic polymerization. | 09 |
| 4(a). Describe the transition state theory of chemical reactions. | 10 |
| 4(b). Write step-by-step free radical mechanism for addition polymerisation by taking ethylene as an example. | 12 |
| 4(c). Write down the importance of microorganism in our daily life. | 07 |
| 4(d). In the photochemical reaction $A \rightarrow B$, 1.00×10^5 mole of B is formed as a result of the absorption of 6.00×10^7 ergs at 3600 Å . Calculate the quantum yield. ($N = 6.023 \times 10^{23}$) | 06 |

SECTION – B

- 5(a). What is threshold value? A combination of SO₂, charcoal powder and high percentage of humidity is the most corrosive environment- explain. 09
- 5(b). Write down the anode and cathode reaction in under-water corrosion. Explain why the presence of acids increases but bases decreases the rate of under-water corrosion. 10
- 5(c). What is passivating agent? Discuss the factors which influence corrosion. 10
- 5(d). Write down the advantage of anodic control of corrosion. 06
- 6(a). Define the following terms giving examples. 09
(i) Adsorption, (ii) Sorption, (iii) Adsorbent.
- 6(b). What is heterogeneous catalysis? Explain the mechanism of heterogeneous catalysis by taking a suitable example. 10
- 6(c). Discuss the effect of temperature on adsorption of gases on solid. 06
- 6(d). Explain why –? 10
(i) Adsorption occurs at the surface, (ii) Finely powdered substance are more effective adsorbents.
- 7(a). Which of the following molecules do not absorb in the infrared region and why? 08
O₂, H₂O, N₂, CO₂
- 7(b). Describe the various types of absorption bands with one example in each case which arise as a result of the electronic transitions. 10
- 7(c). Discuss the mechanism and characteristics of enzyme catalyzed reaction. 10
- 7(d). An organic compound with molecular formula C₂H₆O gives absorption peaks in the regions of 3450 (broad), 2950 – 2850 and 1085 cm⁻¹. Give its probable spectrum and structure. 07
- 8(a). What is normal mode of vibration? How many modes are expected for CO₂ molecule? 10
- 8(b). Calculate the stretching vibration for C=O bond. [Here, mass of C atom = 20×10^{-24} g, mass of O atom = 26.6×10^{-24} g, $k = 5 \times 10^5$ g sec⁻², $C = 3 \times 10^8$ ms⁻¹] 10
- 8(c). Why is methanol a good solvent for UV but not for IR determination? 07
- 8(d). Give an account of finger print region in IR spectra and why is it called so? 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2023

EE 1213

(Electrical Circuits and Electronics)

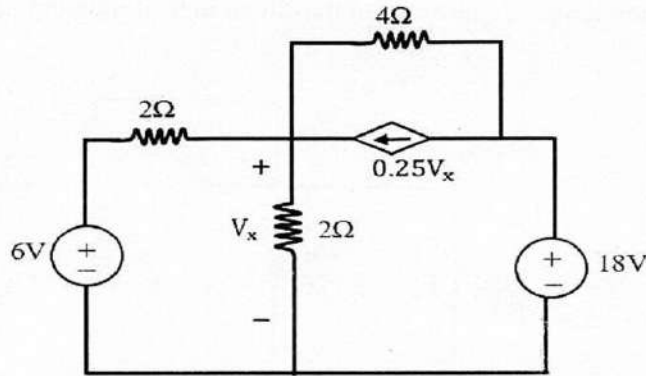
Time: 3 Hours.

Full Marks: 210

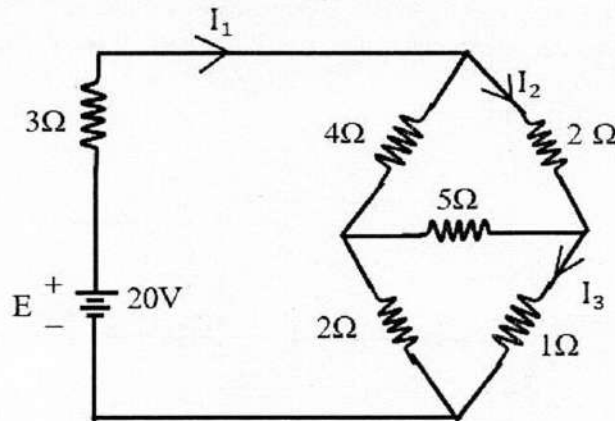
- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define- i) Unilateral network, ii) Bilateral network, iii) Active Components, and iv) Passive Components. 08
- 1(b). Convert a Y-network to an equivalent delta network and find each of the delta-connected resistance in terms of Y-resistance. 10
- 1(c). Calculate V_x in the following figure, using source transformation. 09

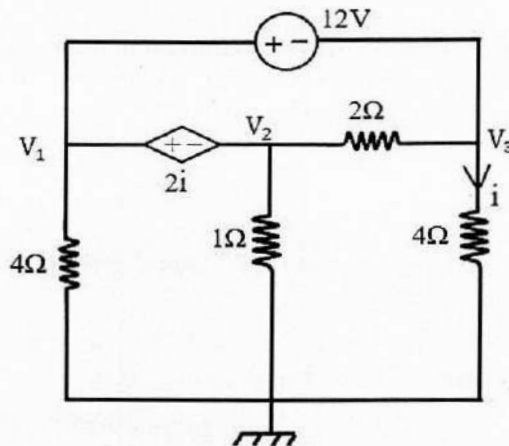


- 1(d). Show that “A load will receive maximum power when the load resistance is equal Thevenin equivalent resistance” Also find the efficiency at this condition. 08
- 2(a). Define electrical sources. Classify and draw the symbols of different electrical sources. 05
- 2(b). Find I_1, I_2 , and I_3 in following figure using mesh-analysis. 10



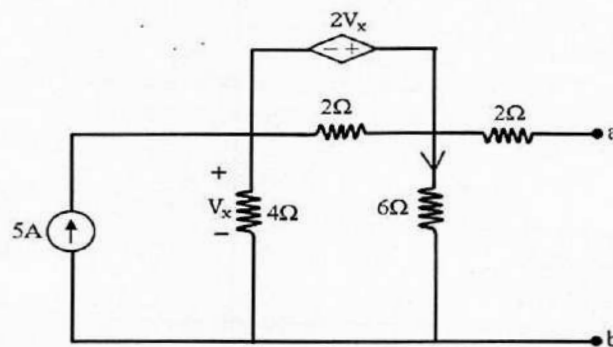
2(c). For the circuit shown in figure, find V_1 , V_2 , and V_3 using Nodal – Analysis.

10



2(d). Find the Thevenin equivalent of the circuit in the following figure at terminals a-b.

10



3(a). Define i) RMS value, ii) Form factor, iii) Power factor.

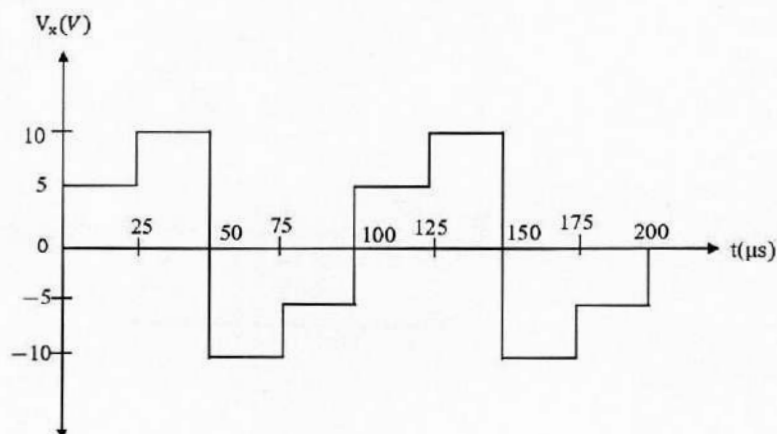
09

3(b). What is phase? Find the angle of phase difference between $V = 100 \cos(\omega t - 30^\circ)$ and $i = -10 \sin(\omega t - 60^\circ)$. Which wave leads than others?

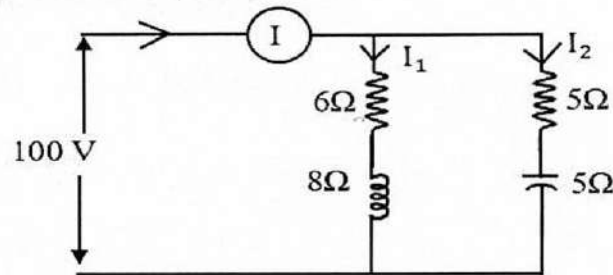
05

3(c). Prove that the effective value of sinusoidal voltage is 70% of the peak voltage. Also determine the power dissipated in a 6Ω resistor, if the voltage shown in figure is applied on it.

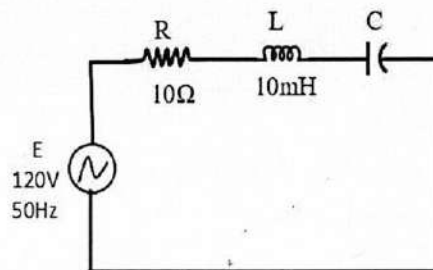
12



- 3(d). What is phasor? Using phasor algebra, find I , I_1 , and I_2 and total power consumed for the following circuit shown in figure. 09



- 4(a). What is resonance? Draw the resonance curve for an R-L-C series circuit. Also find the value of 'C' in figure that will cause resonance at 10^6 Hz frequency. Determine the value of 'Q' too. 15



- 4(b). Can power factor be negative? Explain using impedance and admittance triangle. 08
- 4(c). A RL circuit contains $R = 15\Omega$ and $L = 0.0637$ henry in series, and $i = 10 \cos 157 t$ amp flows through the RL branch. Find the followings: 12
- The equation of voltage v as function of time
 - The expression of power as a function of time
 - The value of impedance of the circuit
 - The amount of energy stored during a quarter cycle.

SECTION – B

- 5(a). Write down the differences among metal, semiconductor and insulator. Why semiconductor acts like an insulator at -273°C ? 12
- 5(b). Derive the following electron concentration equation in CB with proper diagram $n = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$, where the symbols have their usual meanings. 12
- 5(c). Given that the density of states related effective masses of electrons and holes in Si are $1.08 m_e$ and $0.6 m_e$, respectively, and electron and hole drift mobility in room temperature are 1350 and $450 \text{ cm}^2 \text{ v}^{-1} \text{ s}^{-1}$, respectively. Calculate the intrinsic concentration and intrinsic resistivity of Si. 11

- 6(a). What is drift mobility and conductivity? Derive the expression of conductivity of semiconductor material. 08
- 6(b). Show that, the Fermi level in case of n-type material lies near the CB and for p-type material lies near VB. 08
- 6(c). Show that "Depletion region of a p-n junction diode at zero applied bias is proportional to the square root of built in potential (V_0)" 12
- 6(d). Describe the phenomenon: i) avalanche breakdown, and ii) Zener breakdown. 07
- 7(a). Define compensation doping. An n-type Si wafer has been doped uniformly with 10^6 antimony (Sb) atoms cm^{-3} . Calculate the position of the Fermi energy with respect to the Fermi energy E_{Fi} in intrinsic Si. The above n-type Si sample is further doped with 2×10^{17} boron atom cm^{-3} . Calculate the position of Fermi energy with respect to Fermi energy E_{Fi} in intrinsic Si. Assume that $T = 300\text{K}$, $KT = 0.0259\text{ V}$ and $n_i = 1.45 \times 10^{10} \text{cm}^{-3}$. 12
- 7(b). Derive the ideal diode (Shockley) equation $J = J_{so} \left[\exp\left(\frac{eV}{KT}\right) - 1 \right]$, where the symbols have this usual meanings. 14
- 7(c). Draw the energy band diagrams of p-n junction under following conditions: 09
i) Open circuit, ii) Forward bias, iii) Reverse bias.
- 8(a). What is universal gate? Make the basic gates AND, OR, NOT and NOR using NAND gate. 08
- 8(b). Mention undesirable condition of a RS flip-flop. How can we avoid this? Explain. 10
- 8(c). Discuss the operation of a 4-bit shift register with parallel load using necessary diagrams. 10
- 8(d). Design a BCD counter using T-flip flop. 07

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2023

Math 1213

(Differential Equation and Coordinate Geometry)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define invariants. If $4x^2 + 2\sqrt{3}xy + 2y^2 - 1 = 0$ is transformed into $5x^2 + y^2 - 1 = 0$, then find the value of the rotational angle of the latter axes to the former. 10
- 1(b). Find the transformed equation of $x^2 - 4xy + 3y^2 - 1 = 0$ to the new axes X and Y whose equations are $x - y = 0$ and $x + y = 0$, respectively. 11
- 1(c). What conic do the equation $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ represent? Reduce it to standard form. 14
- 2(a). Find the cylindrical polar coordinates and spherical polar coordinates whose cartesian coordinates are $(-1, -\sqrt{3}, -2)$. 12
- 2(b). Find the ratio in which the zx -plane divides the segment joining the points $(1, 3, -5)$ and $(2, -4, 1)$. Also find the intersection point of this line with the zx -plane. 11
- 2(c). Find the angle between any two diagonals of a cube. 12
- 3(a). Derive the symmetric or standard form of the equation following line: $3x + 2y - z - 4 = 0 = 4x + y - 2z + 3$
Also find its direction cosines. 12
- 3(b). Determine the shortest distance between two skew-lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ 13
- 3(c). What is plane? Find the angle between the line $\frac{x+1}{3} = \frac{y-1}{-6} = \frac{z-1}{-6}$ and the plane $2x - 2y + z - 3 = 0$. 10
- 4(a). Find the vector equation of a plane passing through the point $(1, -2, 3)$ and having the direction ratio of its normal are 5, 2, 4. 12
- 4(b). Find the equation of a line in vector form which passes through the point $(-1, 2, 1)$ and parallel to the line $\frac{x-5}{2} = \frac{y-2}{4} = \frac{z-7}{3}$. 11
- 4(c). A line passes through the points A $(1, 0, 2)$ and B $(3, 9, 6)$. Find the equation of this line in (i) Vector form, (ii) Parametric form, and (iii) Symmetric form. 12

SECTION – B

- 5(a). Define general solution of an ordinary differential equation. 12
Solve the D.E. $(x + y)dx - xdy = 0$
- 5(b). Discuss $\left(y + \frac{y^3}{3} + \frac{x^2}{2}\right)dx + \frac{1}{4}(x + xy^2)dy = 0$ is exact or not. 12
If it is not exact, then determine the general solution of it.
- 5(c). A body of temperature 80°F is placed in a room of constant temperature 50°F at time $t = 0$. At the end of 5 minutes, the body has cooled to a temperature of 70°F . 11
(i) Determine the temperature of the body as a function of time $t > 0$.
(ii) Find the temperature of the body at the end of 10 minutes.
- 6(a). A circuit has in series an electromotive force given by $E = 100V$, a resistor of 10Ω and inductor of $0.5H$. If the initial current is zero, then find the current at time $t > 0$. 12
- 6(b). Solve: $(D^2 + 2D + 1)y = xe^{5x}$. 11
- 6(c). Solve: $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 8e^{2x} - 5e^{3x}$; $y(0) = 3, y'(0) = 5$ 12
- 7(a). An 8-lb weight is placed upon the lower end of a coil spring suspended from the ceiling. The weight comes to rest in its equilibrium position, thereby stretching the spring 6 inches. The weight is then pulled down 3 inches below its equilibrium position and released at $t = 0$ with an initial velocity of 1 ft/sec, directed downward. Neglecting the resistance of the medium and assuming that no external forces are present, determine the amplitude, period, and frequency of the resulting motion. 14
- 7(b). Solve $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$ by the method of variation of parameters. 11
- 7(c). Define Laplace transform. Find the Laplace transform of $G(t)$, where 10

$$G(t) = (t - 2)^3 \text{ for } t > 2$$

$$= 0 \text{ for } t < 2$$
 by using second shifting property.
- 8(a). Find the inverse Laplace transform of $\frac{3s-14}{s^2-4s+8}$ 12
- 8(b). Find the inverse Laplace transform of $\frac{s}{(s^2+3^2)^2}$ by using Convolution theorem. 10
- 8(c). Solve: $\frac{d^2Y}{dt^2} + 9Y = \cos 2t$; $Y(0) = 1, Y\left(\frac{\pi}{2}\right) = -1$ by using Laplace transform. 13

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2023

Hum 1213

(Technical English)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Make sentences with the following structures using the words given in brackets. 14
- i) Subj. + Transitive verb + Object. (Feel as verb)
 - ii) Subj. + Intransitive verb + Adv. of Manner. (Sleep as verb)
 - iii) That + Subj. + Verb + Adv. of Manner + Verb + Adj. complement. (Behave and is as verb)
 - iv) Subj. + Verb + Not only + Obj. + But also + Obj. (Read as verb)
 - v) Subj. + Relative pronoun + Verb + Adv. + Verb + Adj. complement. (Exercise and is as verb)
 - vi) Though + Subj. + Verb + Adv., subj. + Verb + Obj. (is and like as verb)
 - vii) Subj. + Verb + as + Subj. + Verb. (Do and teach as verb)
- 1(b). Change the following words as asked in brackets and use the changed forms in sentence. 12
- i) Acuity (into adjective), ii) Impiety (into adjective), iii) Lively (into noun), iv) Elude (into noun), v) Palpation (into verb), vi) Abandon (into noun).
- 1(c). Make words with each of the following suffixes and prefixes and use them in sentence. 09
- fold, —tic, —fy, dys—, cu—, tri—.
- 2(a). Transform the following sentences as asked in brackets. 14
- i) Christiana, a banker, works in Khulna. (Complex)
 - ii) Habib is performing duties is praiseworthy. (Complex)
 - iii) As Didar can work hard, he can complete courses in time. (Simple)
 - iv) In spite of Nasim's idleness he can do well in duties. (Complex)
 - v) Mim is as liberal as his friends. (Comparative)
 - vi) Ohid walks fast, so can reach in class in time. (Complex)
 - vii) At boyhood Paul obeyed his parents. (Complex)
- 2(b). Make use of the following words in sentence as asked in brackets. 12
- Calm (as verb), Face (as verb), Mind (as verb), Above (as adj.), Hold (as noun), Man (as verb)
- 2(c). Write one antonym and one synonym for the words given below and use the antonym and synonym in sentence. 09
- i) Free, ii) Intimate, and iii) Wise.

- 3(a). Make WH questions with the underlined words of the following sentences. 14
- Habib delivered us a fine speech on sacrifice.
 - The operation lasted an hour and a half.
 - He walks five miles everyday.
 - The building is two and a half metre high.
 - Liza endeavors to succeed in life.
 - Shila moves with a rapid speed to reach the class
 - The poem is highly revolutionary.
- 3(b). Complete the sentences with subordinate clauses: 12
- _____ is true. (Noun clause)
 - It depends on _____. (Noun clause)
 - This is the man _____. (Adj. clause)
 - The gallery _____ was crowded. (Adj. clause)
 - _____, he is victim. (Adv. clause of extent)
 - They started early _____. (Adv. clause of purpose)
- 3(c). Express the following notions and functions in sentences: 09
- Determination,
 - Hopefulness,
 - Hatred,
 - Congratulations,
 - Proposal,
 - Request.
- 4(a). Correct the following sentences: 14
- Chairman will preside over the meeting.
 - The sunshine is essentially beneficial to our body.
 - Mamun's younger is punctual.
 - Hasim repairs his furnitures.
 - Kamal is sick.
 - We say him a liar.
 - Hanna did a mistake.
- 4(b). Make use of the following modal auxiliaries in sentence as asked in brackets. 12
- Can. (To express an offer to somebody else)
 - Can. (To express ability)
 - Would. (To express a past irregular habit)
 - Should. (To express propriety)
 - Would rather. (To express preference)
 - Will. (To express a polite request)
- 4(c). Make sentences with the following idioms and phrases: 09
- (i) Walk on eggshells, (ii) Make two ends meet, (iii) Clouds on the horizon, (iv) Beating a dead horse, (v) Worth its weight in gold, (vi) Like two peas in a pod.

SECTION – B

- 5(a). Read the passage and answer the following questions. 20
- One morning Savni received a parcel from America. She supposed it was from her Aunt Pasni. As she didn't hear anything from her in the last twenty years, she was very surprised to have the parcel. Setting the parcel on the table, Savni carefully began to tear open the paper. Inside, there was a large silver container and a letter. Savni and her family had no idea what it was. So she opened the letter, then looked up in dismay. "This is no help! It's written in English- we are poor people with no education," said Savni. Cautiously, Savni pulled the lid of the pot. All they could see is powder. They were all mystified. Muma said it must be American coffee but her father rejected the idea as the powder was light-grey colored. Avinas, dipping a finger into the pot, licked it and said, "it's mild but quite tasty. It's America pepper." Savni said, "we'll try it on the stew tonight and have American style stew!"
- That evening, the whole family agreed that the American pepper had added a special extra taste to their usual evening stew. They were delighted with it. By the end of the week, there was only a teaspoonful of the grey powder left. So Savni caved it for Dr. Haret as a special treat.
- The following Sunday, the whole family was ready for dinner with Dr. Haret. During dinner, Savni explained to the doctor about the mysterious American pepper and the letter.
- "Well, give it to me, I speak English! I will translate it for you."
- The family waited fascinatedly as the doctor translated: "Dear Savni: I am the son of your Aunt Pasni. In her final days of illness, she told us that after her death, she wanted her ashes to be sent to you, so that you could scatter them on the hills of the country where she was born. My mother died last week. After her cremation, I am sending her ashes to you in a silver casket. Please do as asked. Your cousin, George Leary."
- (i) Why did the family think that the gray powder was pepper?
- (ii) How did Dr. Haret solve the mystery?
- (iii) What is your opinion regarding Savni and her family consuming the powder in stew?
- (iv) If you were Savni, how would you respond to a mysterious parcel?
- 5(b). Make a précis of the above passage mentioned in question no. 5(a) with a title. 15
- 6(a). Write a cause and effect paragraph on procrastination. 15
- 6(b). Amplify the idea – 'Tolerance of intolerance is cowardice'. 20
- 7(a). Suppose you have completed your B.Sc. Engineering want to join some project running in your department. Prepare your CV as a research assistant and apply to the project manager. 15
- 7(b). It came to your attention that many employees are using unplanned non-working hours after their scheduled lunch breaks. Write a memo introducing a concept to involve stapes into productive activities to reduce non-working hours. 20
8. Write a free composition on any one of the followings: 35
- i) The world and climate change: So called developments
- ii) Social Identity.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2023

ESE 1205

(Thermodynamics for Energy Engineering)

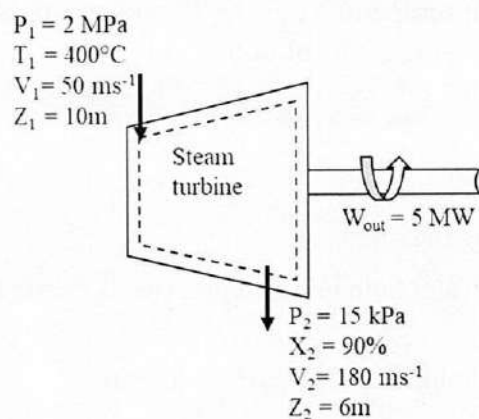
Time: 3 Hours.

Full Marks: 210

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SECTION – A

- 1(a). Consider (i) a mixture of liquid and gaseous water, and (ii) a mixture of liquid and gaseous air. Which one is pure substance? Explain. 08
- 1(b). What is state postulate? Write down Gibb's phase rule for non reacting system and mention the meaning of each term. 08
- 1(c). Write short note on (i) Quasi-equilibrium process, (ii) Intensive property, (iii) Thermodynamic equilibrium, (iv) Thermodynamic system, (v) External property. 10
- 1(d). Determine the pressure exerted by 4.0 kg CO in a 0.03 m³ container at 220 K by using ideal gas law, vander Wal's and redlic-Kwong equation. 09
- 2(a). State first law of thermodynamics. Draw T-v diagram of a single component pure substance and label constant pressure line, saturation line, critical point, regions. 10
- 2(b). Write down the energy balance equation for a general steady flow system and describe what each term in the equation means. 10
- 2(c). The power output of an adiabatic steam turbine is 5 MW and the inlet and exit condition of the steam are indicated in the following figure: 15



- (i) Compare the magnitudes of Δh , ΔK_e , and ΔP_e
- (ii) Determine the work done per unit mass of the steam flowing through the turbine.
- (iii) Calculate the mass flow rate of steam.
- 3(a). Write down the characteristics of heat engine. Also discuss the thermal efficiency of heat engine. 07
- 3(b). "The efficiencies of all reversible heat engines operating between the two reservoirs are the same" – Prove the statement. 10
- 3(c). Air is compressed by an adiabatic compressor from 100 kPa and 12°C to a pressure of 800 kPa at a steady rate of 0.2 kg/s. If the isentropic efficiency of the compressor is 80%, then determine (i) the exit temperature of air, (ii) the required power input to the compressor. 18
- 4(a). Explain the significance of Maxwell relations in thermodynamics. How do they allow for the determination of entropy changes in a system where direct measurement is not possible? 12
- 4(b). Write short note on (i) Volume Expansivity, (ii) Isothermal Compressibility. 08
- 4(c). Show that exergy of a flow stream is 15

$$X = (h - h_0) - T_0(s - s_0) + \frac{v^2}{2} + gz$$

Where the symbols have their usual meaning.

SECTION – B

- 5(a). Write down the classifications of thermodynamic cycle. Also discuss the processes of Carnot cycle with necessary diagrams. 10
- 5(b). Describe different methods of increasing the efficiency of the Rankine cycle with necessary diagrams. 12
- 5(c). Consider a steam power plant operating on ideal reheat Rankine cycle. Steam enters the high-pressure turbine at 15 MPa and 600°C and is condensed in the condenser at a pressure of 10 kPa. If the moisture content of the steam at the exit of the low-pressure turbine is not exceed 10.4%, determine (i) the pressure at which the steam should be reheated, and (ii) the thermal efficiency of the cycle. Assume the steam is reheated to the inlet temperature of high-pressure turbine. 13
- 6(a). Derive the expression of the efficiency for an ideal regenerative Rankine cycle. 12
- 6(b). Differentiate between open and closed feed water heaters. 08
- 6(c). A gas turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Utilizing the air-standard assumptions, determine (i) the gas temperature at the exit of the compressor and the turbine, (ii) the thermal efficiency. 15
- 7(a). Differentiate between vapor compression refrigeration system and vapor absorption refrigeration system. 08
- 7(b). Describe the working principle of a four-stroke Otto cycle with necessary diagrams. 12
- 7(c). In an air standard diesel cycle, the compression ratio is 16 and at the beginning of isentropic compression, the temperature is 15°C and the pressure is 0.1 MPa. Heat is added until the temperature at the end of constant pressure is 1480°C. Calculate (i) the cutoff ratio, (ii) heat supplied per kg of air, (iii) the cycle efficiency, and (iv) the mean effective pressure. 15
- 8(a). Write short note on the followings: 10
(i) Dry bulb temperature, (ii) Wet bulb temperature, (iii) Specific humidity, and (iv) Relative humidity.
- 8(b). Derive the expression for calculating degree of saturation. 10
- 8(c). Draw and mention different parameters of a psychrometric chart. 05
- 8(d). A mixture of ideal gases consists of 3 kg of nitrogen and 5 kg of carbon dioxide at a pressure of 300 kPa and a temperature of 20°C. Calculate (i) mole fraction of each constituent, (ii) equivalent molecular weight of the mixture. 10

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